

Freight Rate Prediction

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\$120.87	\$637.53	0.8255	50.9%
HELD-OUT MAE	HELD-OUT RMSE	HELD-OUT R-SQUARED	MAE GAIN VS BEST BASELINE

Executive summary

The task is a forward forecasting problem: 48,000 labeled loads cover January-October 2025, while 12,000 unlabeled loads cover November-December. The solution uses forward-only temporal validation, training-partition-specific cleaning, engineered route and calendar features, and a tuned LightGBM regressor. Both required prediction files passed the supplied structural scorer.

Temporal validation design

Stage	Training period	Evaluation period	Purpose
Inner fold 1	January-May	June	Tuning and early stopping
Inner fold 2	January-June	July	Tuning and early stopping
Inner fold 3	January-July	August	Tuning and early stopping
Held-out assessment	January-August	September-October	One-time evaluation after settings were frozen

A random split was avoided because it would mix later market conditions into training and overstate performance for the actual future prediction task.

Data-quality treatment

Issue	Treatment	Leakage control
Negative weight	Use absolute magnitude as a sign-entry correction.	Feature-only correction applied consistently.
Missing values	Retain for native model handling; avoid arbitrary constants.	No target-based imputation.
Extreme log-RPM	Exclude $ z > 3.5$ observations from fitting only.	Recalculate statistics within each training partition; never filter evaluation rows.

Held-out September-October performance

Method	MAE	RMSE	R-squared
Median RPM x distance	\$256.95	\$684.24	0.7990
Distance x quote signal	\$246.02	\$711.80	0.7824
Selected LightGBM	\$120.87	\$637.53	0.8255

Fixed December prediction chart

Required fixed lane: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb

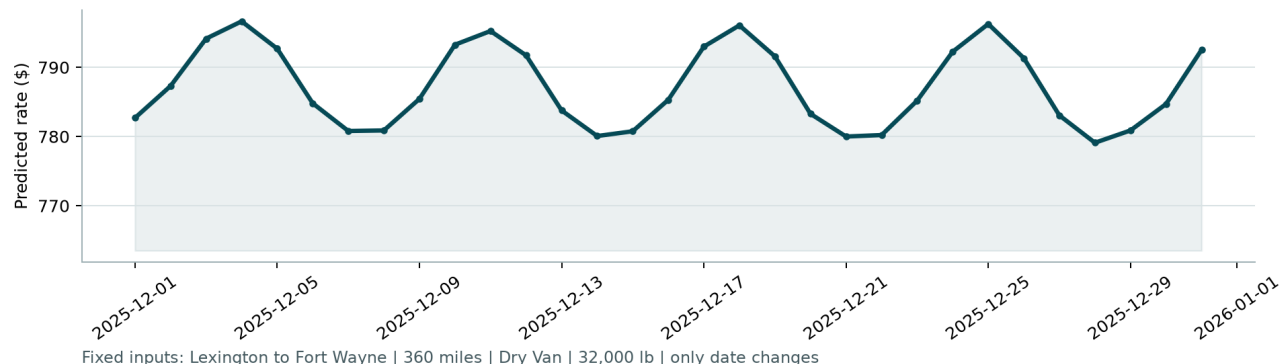


Chart construction and assumptions

Element	Implementation
Fixed visible inputs	Pickup, delivery, distance, equipment, and weight remain unchanged for all 31 days.
Coordinates	Obtained from the consistent city-coordinate mapping in the development data.
Daily market signals	The chart input file does not include market_index or quote_signal, so their daily averages were taken from validation.csv and matched by date.

Final pipeline

Model fitting LightGBM on route, shipment, location, market, and cyclic calendar features; frozen hyperparameters and 2,452 trees.	Required outputs 12,000 load-level predictions and 31 fixed-lane December predictions.
Validation All IDs unique and complete; all predicted rates finite and positive.	Scorer check Both required output files passed the supplied structural validation checks.

Conclusion

The final model was used to predict rates for all 12,000 November-December loads and to generate the fixed Lexington-to-Fort Wayne December chart.